Assignment No:9

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Batch:A

Title:Develop an AI model to analyze social media sentiment trends for financial markets.

```
#Import Libraries
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt

from sklearn.model_selection import train_test_split
from sklearn.feature_extraction.text import TfidfVectorizer
from sklearn.linear_model import LogisticRegression
from sklearn.metrics import accuracy_score, classification_report
```

```
#Load Your Dataset
df = pd.read_csv("data.csv")

print(df.head())
print(df.columns)
```

```
                                            Sentence Sentiment
0  The GeoSolutions technology will leverage Bene...  positive
1  $ESI on lows, down $1.50 to $2.50 BK a real po...  negative
2  For the last quarter of 2010 , Componenta 's n...  positive
3  According to the Finnish-Russian Chamber of Co...   neutral
4  The Swedish buyout firm has sold its remaining...   neutral
Index(['Sentence', 'Sentiment'], dtype='object')
```

```
#Data Preprocessing
# Rename columns if needed
df.columns = df.columns.str.lower()

# Example (adjust if names differ)
text_col = 'text' if 'text' in df.columns else df.columns[0]
label_col = 'sentiment' if 'sentiment' in df.columns else df.columns[-1]

# Remove null values
df = df[[text_col, label_col]].dropna()

print(df.head())
```

```
                                            sentence sentiment
0  The GeoSolutions technology will leverage Bene...  positive
1  $ESI on lows, down $1.50 to $2.50 BK a real po...  negative
2  For the last quarter of 2010 , Componenta 's n...  positive
3  According to the Finnish-Russian Chamber of Co...   neutral
4  The Swedish buyout firm has sold its remaining...   neutral
```

```
#Convert Text → Features (TF-IDF)
vectorizer = TfidfVectorizer(stop_words='english', max_features=5000)

X = vectorizer.fit_transform(df[text_col])
y = df[label_col]
```

```
#Train Model
X_train, X_test, y_train, y_test = train_test_split(X, y, test_size=0.2)
```

```
model = LogisticRegression()
model.fit(X_train, y_train)
```

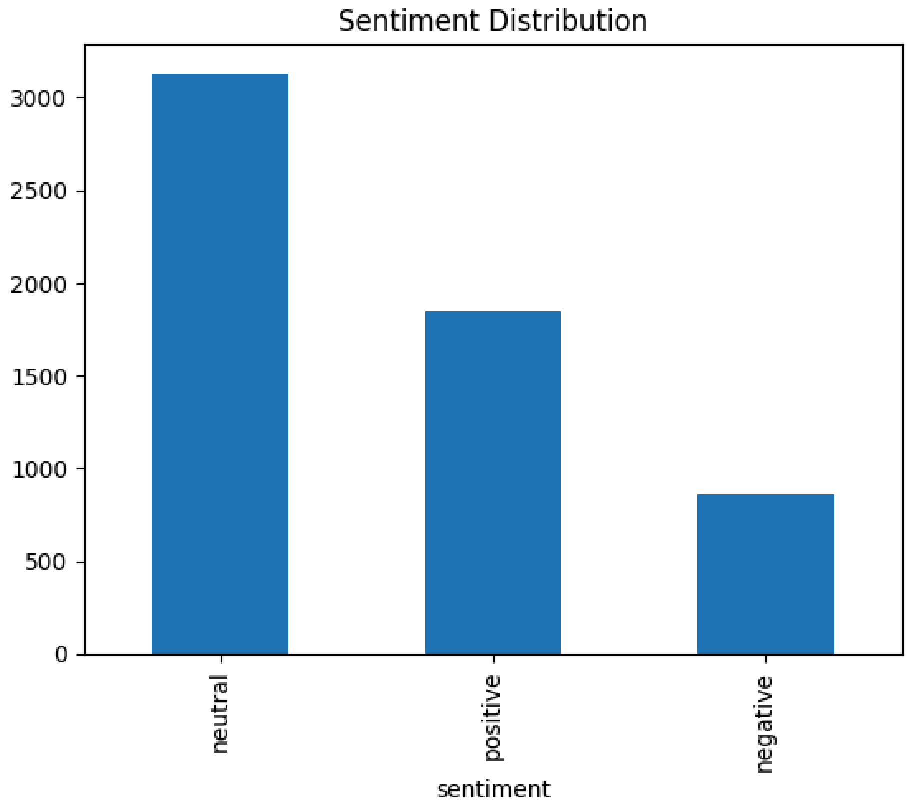
```
▾ LogisticRegression
LogisticRegression()
```

```
#Evaluate Model
y_pred = model.predict(X_test)

print("Accuracy:", accuracy_score(y_test, y_pred))
print(classification_report(y_test, y_pred))
```

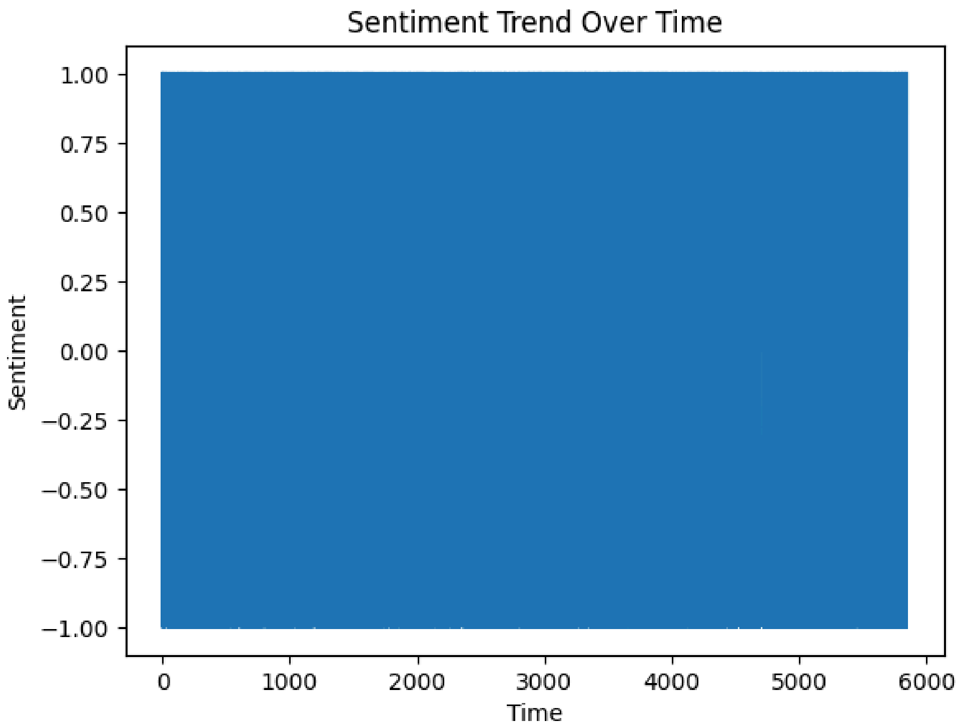
```
Accuracy: 0.6911890504704876
              precision    recall  f1-score   support

    negative       0.47      0.16      0.24       189
     neutral       0.68      0.91      0.78       616
    positive       0.77      0.60      0.67       364

    accuracy                           0.69      1169
   macro avg       0.64      0.56      0.56      1169
weighted avg       0.67      0.69      0.66      1169
```

```
#Convert sentiment to numeric
sentiment_map = {
    'positive': 1,
    'neutral': 0,
    'negative': -1
}

df['sentiment_score'] = df[label_col].map(sentiment_map)
```

```
#Plot Sentiment Distribution
df[label_col].value_counts().plot(kind='bar')
plt.title("Sentiment Distribution")
plt.show()
```




```
#Sentiment Trend (Simulated Time Series)

df['index'] = range(len(df))

plt.plot(df['index'], df['sentiment_score'])
plt.title("Sentiment Trend Over Time")
plt.xlabel("Time")
plt.ylabel("Sentiment")
plt.show()
```




```
#Financial Insight (Basic Simulation)
```

```
avg_sentiment = df['sentiment_score'].mean()
print("Average Market Sentiment:", avg_sentiment)
```

```
Average Market Sentiment: 0.16980486134885314
```

```
#Predict New Tweet Sentiment
def predict_sentiment(text):
    vec = vectorizer.transform([text])
    pred = model.predict(vec)
    return pred[0]

print(predict_sentiment("Stock market is going up strongly"))
```